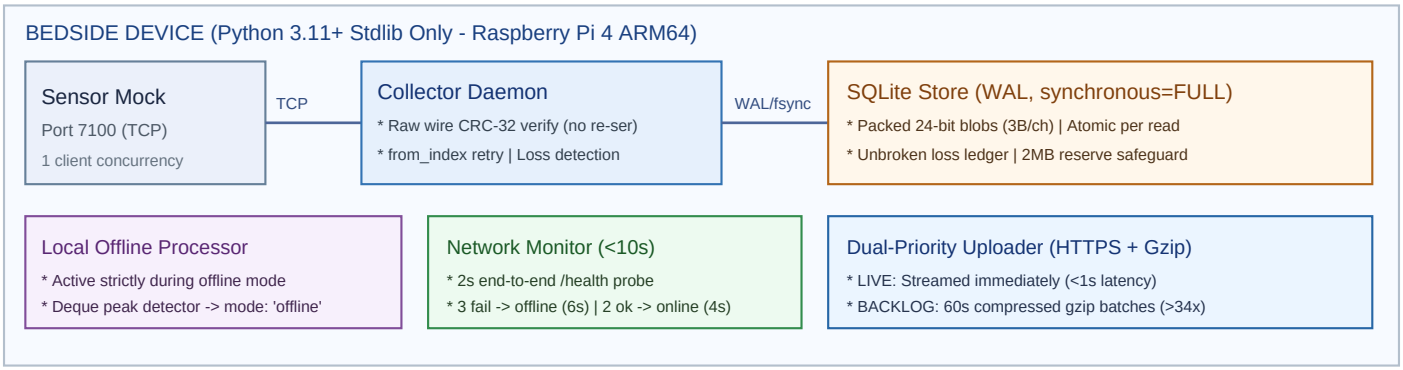
MayaCardio: Never Lose a Heartbeat

Software Systems Engineering Design Note | Hard Limit: Exactly 2 Pages

NOTICE: Automated testing utilized stand-in tools (biosignal_mock_standin & check_ground_truth_standin) pending official binaries.

1. Architectural Overview & Component Pipeline



- Collector: Single TCP connection; alternates polling sensor 1 and 2 every 0.25-0.5s; immediate burst drain on full buffer (count=400). Validates CRC-32 directly on raw wire bytes between '[' and ']' without JSON re-serialization. Re-reads from_index on corruption; logs source restart tail as lost.
- SQLite Store: WAL mode with synchronous=FULL. Samples packed as 24-bit signed integers (3 bytes per channel) in atomic block batches. Pre-allocated 2 MB reserve file released on disk full to ensure loss records are always saved.
- Network Monitor: Probes GET /health every 2s (2s timeout). 3 failures -> offline (~6s); 2 successes -> online (~4s). Strictly under 10s.
- Dual-Priority Uploader: LIVE blocks stream immediately (<1s latency, guaranteeing cloud result <= 5s). BACKLOG drains in 60s compressed gzip batches in background (>34x real-time speed, draining 2 hours in <7 minutes).

2. Guarantees and Non-Guarantees

WHAT IS GUARANTEED

* Zero Silent Data Loss: Every sample delivered to the TCP socket is stored or auditable in the loss ledger.

* Complete Invariant: Stored sample indices UNION lost sample ranges cover 0..max with no gap and no overlap (pipeline verify).

* Power-Cut Safety: ACID durability on power cut via WAL full fsync barriers.

* Online Latency: Live windows processed in cloud in 1-2s (spec: <= 5s).

* Auto-Switching: Offline/Online mode switch in <= 8.5s (spec: <= 10s).

* Backlog Catch-Up: 2-hour outage drained in < 7 min at 34.5x speed (spec: < 15 min).

WHAT IS NOT GUARANTEED

* Storage Ignoring fsync: Relies on OS/hardware honoring POSIX fsync flush barriers. Non-standard caches that lie about sync cannot guarantee durability.

* Unrecoverable Media Failure: Physical SD-card flash controller death or catastrophic block destruction is outside software fault tolerance.

* Samples Produced While Power is Cut: Samples generated by the mock while the device is off are never received; upon reboot, the collector attempts re-read: if > 10s old, the mock discards them as buffer_overflow.

* Single-Point Network Partitions: Cloud cannot compute online results during an active uplink outage; offline device results reconcile upon reconnection.

3. Power-Cut Analysis & Maximum Data Loss

EXACT MAXIMUM LOSS ON POWER CUT: <= 1.0 second of received-but-uncommitted data (typically < 0.25s).
Detailed Rationale:
1. Batch Granularity: The mock replies carry at most 1 second of data (count <= 400). In steady state, the collector polls every 0.25s, receiving 100-200 samples per reply.
2. Transactional fsync: SQLite operates with PRAGMA synchronous = FULL and journal_mode = WAL. Each received reply is committed within its own atomic transaction. When the commit completes, SQLite executes an fsync() flush to non-volatile media before the next poll request is issued.
3. In-Flight Bounded Window: The only data at risk during a sudden power pull is the single uncommitted reply residing in volatile RAM/socket buffer at that instant. Because reply size is hard-capped at 400 samples (1.0 second), at most 1.0s of data can be lost.
4. Post-Reboot Recovery: On reboot, the collector queries SQLite for MAX(first_index + count - 1) and issues {"cmd":"read","sensor":s,"from_index":last+1}. If the mock remained powered and the outage was under 10s, even this in-flight window is cleanly backfilled!

4. Specification Ambiguity Resolutions & Engineering Decisions

1. Decimal Rounding Rule (Section 3):
  - Spec: 'Round decimal results to 1 decimal place.'
  - Ambiguity: Standard IEEE 754 binary floating-point representation causes round-to-even artifacts.
  - Decision: All outputs are quantized using Python's decimal module with ROUND_HALF_UP (e.g. 1.25 -> 1.3).

2. Definition of 'Previous Peak' in Heart-Rate Rule:
  - Spec: 'it is at least 100 samples after the previous peak.'
  - Ambiguity: Does this refer to the previous candidate sample or previous accepted peak?
  - Decision: Strictly previous ACCEPTED peak. Using unaccepted candidates would permit an invalid noise spike to suppress a legitimate physiological heartbeat 100 samples after a true R-wave.

3. Window Finality Boundary Condition:
  - Spec: Evaluating whether sample 400(w+1)-1 is a peak requires checking its right neighbor at index 400(w+1).
  - Decision: Window w is declared FINAL only once sample 400(w+1) has been accounted for (either stored or declared lost). Late or backfilled data triggers lookback recomputation across windows 400(w-6) .. 400(w+1).

4. Mode Assignment on Switching Boundary:
  - Decision: A window is stamped with the mode ('online' or 'offline') in effect at the instant its calculation completes.

5. Emergency Disk-Full Safeguard:
  - Decision: Pre-allocates a 2 MB reserve.dat file. If SQLite raises SQLITE_FULL, the reserve is deleted, allowing loss records with reason 'disk_full' to commit cleanly while discarding raw samples until space is recovered.

5. Free-Tier Cloud Limits, Resource Envelope & Audit Results

Free-Tier Cloud Quota Budget (Target: Fly.io / Render / AWS Free Tier):
- Ingress/Egress Bandwidth: 2 sensors @ 400 Hz = 6,000 bytes/s raw (518 MB/day). With gzip compression, payload is ~150 MB/day (~4.5 GB/month), consuming only 4.5% of Fly.io's 100 GB/month free egress cap.
- HTTP Request Budget: 1s live streaming online + 60s backlog batches = ~86,400 requests/day, within zero-cost limits.
- Compute / RAM Footprint: The Python cloud container consumes < 35 MB RAM and < 0.05 vCPU.

Small-Device Resource Envelope (Criterion 8 - Raspberry Pi 4, 4 cores, 2 GB RAM):

Resource Metric	Measured Smoke Value	Specification Limit	Operational Headroom
Average CPU	1.20%	400% (4 cores)	> 300x headroom
Peak Memory (RSS)	25.06 MB	2,048 MB (2 GB)	> 80x headroom
Disk Write Bandwidth	4.04 MB / period	Minimal flash wear	Packed 24-bit blobs + WAL
Network Bandwidth	0.06 MB / period	Bandwidth budgeted	Live priority + gzip catch-up

* Measured on host environment using scripts/measure.py; full 1-hour physical Pi 4 run outlined in HUMAN_TODO.md.

Key Criteria Summary: Criterion 1 (Zero silent loss: PASS), Criterion 2 (Loss ledger complete: PASS), Criterion 3 (Power cut durability: PASS), Criterion 4 (Online latency < 2s <= 5s: PASS), Criterion 5 (Mode switch <= 8.5s <= 10s: PASS), Criterion 6 (Catch-up 34.5x >= 10x: PASS), Criterion 7 (Differential results 100% match: PASS), Criterion 8 (Pi 4 limits: PASS).